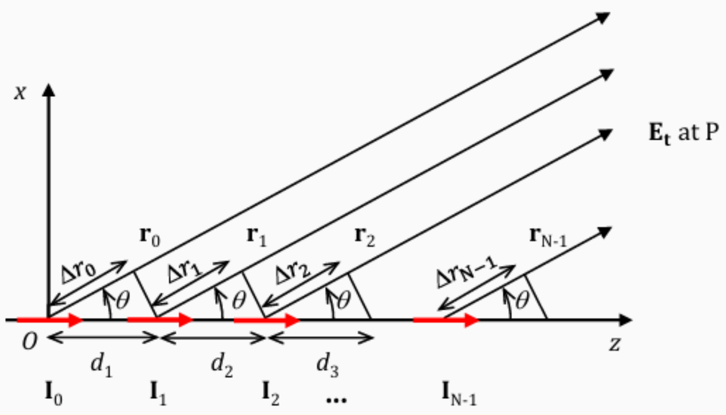
Array Factor



N identical Hertzian dipoles with a length l in z-axis

- Linearly inter-element spaced in z-direction: $d_1, d_2, d_3, \ldots, d_{N-1}$
- Distance between elements and point P in x-z plane: $\mathbf{r}_0, \mathbf{r}_1, \ldots, \mathbf{r}_{N-1}$
- Excitation currents on elements: $\mathbf{I}_n = \mathbf{a}_z I_n e^{jn\beta}$

β: progressive phaseshift from $\mathbf{I}_0$

Uniform Spaced and Phased Linear Array

$0 < \theta < \pi$

$\psi = kd\cos\theta + \beta$

$$AF(\theta) = \sum_{n=0}^{N-1} \frac{I_n}{I_0} e^{jn(kd\cos\theta + \beta)} = \sum_{n=0}^{N-1} \frac{I_n}{I_0} e^{jn\psi}$$

$$|AF(\theta)|_{max} = \left|\sum_{n=0}^{N-1} \frac{I_n}{I_0}\right|$$

Uniform Spaced, Phased and Current Amplitude Linear Array

$$AF(\theta) = \sum_{n=0}^{N-1} e^{jn(kd\cos\theta + \beta)} = \sum_{n=0}^{N-1} e^{jn\psi}$$

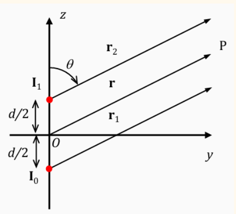
$$|AF(\theta)|_{max} = N \quad if\ \psi = 0$$

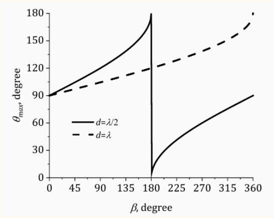
$$AF(\theta) = \frac{1 - e^{jN(kd\cos\theta + \beta)}}{1 - e^{j(kd\cos\theta + \beta)}} = \frac{1 - e^{jN\psi}}{1 - e^{j\psi}}$$

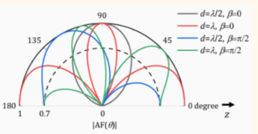
$$|AF(\theta)| = \left|\frac{\sin(N(kd\cos\theta + \beta)/2)}{\sin((kd\cos\theta + \beta)/2)}\right| = \left|\frac{\sin(N\psi/2)}{\sin(\psi/2)}\right|$$

$$\beta = -kd\cos\theta_{max}$$

Array Factor 2 Elements (if unclear β starts counting from which element, go from 1st principles)







	Mainbeam direction	Null direction	HPBW
$d=\lambda/2, \beta=0$	90°	0°, 180°	60°
$d=\lambda, \beta=0$	0°, 90°, 180°	60°, 120°	82.8°, 29°, 82.8°
$d=\lambda/2, \beta=\pi/2$	120°	60°	90°
$d=\lambda, \beta=\pi/2$	104.5°, 41.4°	138.6°, 75.5°	30°, 60°

(b) Spacing d and phase shift β

- θ_{max} points to broadside (θ=90°) as β=0°
- θ_{max} shifts from broadside as β increases from 0°
- Larger kd narrows the range of θ_{max} because of more sidelobes
  - $d=\lambda/2$: $0° \le \theta_{max} \le 180°$
  - $d=\lambda$: $90° \le \theta_{max} \le 180°$

*Importantly, all directions must be $0° \le \theta \le 180°$ as n is selected in calculation.

$0 \le \theta \le \pi$

$$AF(\theta) = 2\cos\left(\frac{1}{2}kd\cos\theta + \frac{\beta}{2}\right)$$

$$MAX\{|AF(\theta)|\} = 2$$

$$\theta_{max} = \cos^{-1}\left(\frac{1}{kd}(-\beta \pm 2n\pi)\right) \qquad n = 0, 1, 2, \ldots$$

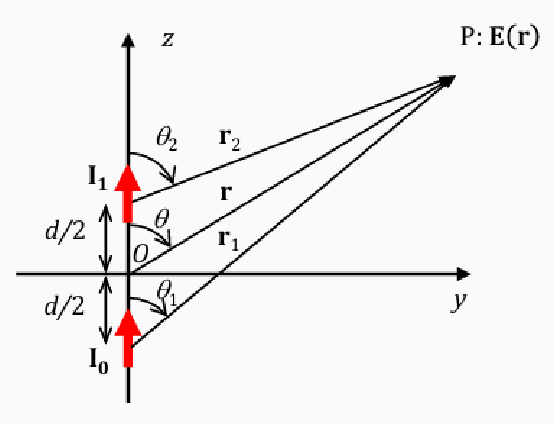
$$\theta_{HPBW} = \cos^{-1}\left(\frac{1}{kd}\left(-\beta \pm 2\cos^{-1}\left(\sqrt{1/2}\right)\right)\right)$$

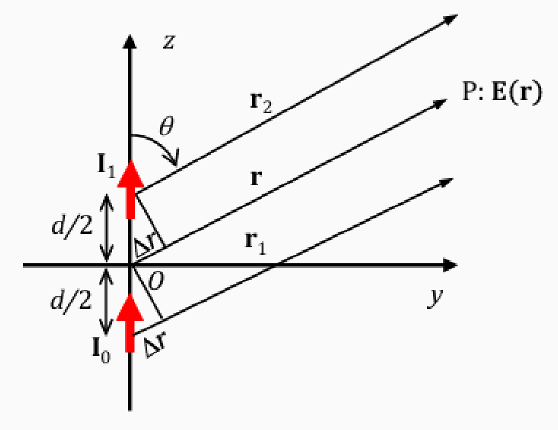
$$HPBW = \theta_{HPBW2} - \theta_{HPBW1}$$

$$|AF(\theta_{null})| = 0$$

$$\theta_{null} = \cos^{-1}\left(\frac{1}{kd}(-\beta \pm (2n+1)\pi)\right) \qquad n = 0, 1, 2, \ldots$$

Element Factor





Two ideal dipoles with length l with spacing d, oriented in z-direction, positioned in z-axis, and centered at origin O, carrying currents $\mathbf{I}_1=\mathbf{a}_z I_1 e^{-j\frac{\beta}{2}}$ and $\mathbf{I}_2=\mathbf{a}_z I_2 e^{j\frac{\beta}{2}}$ and phase difference β. Observation point P is located in y-z plane and in farfield.

1. Approximation of distance:
   $\mathbf{r}_1 || \mathbf{r}_2 || \mathbf{r}$ for $\mathbf{r}_1=\mathbf{a}_r r_1$, $\mathbf{r}_2=\mathbf{a}_r r_2$, and $\mathbf{r}=\mathbf{a}_r r$
   For amplitude: $r \approx r_1 \approx r_2$
   For phase: $r_1=r+\Delta r=r+\frac{d}{2}\cos\theta$ and $r_2=r-\Delta r=r-\frac{d}{2}\cos\theta$
2. Radiated field from $\mathbf{I}=\mathbf{a}_z I$ located at origin O:
$$\mathbf{E}_0(\mathbf{r},\theta)=\mathbf{a}_\theta j\eta k I l \frac{e^{-jkr}}{4\pi r}\sin\theta$$
3. Total radiated field is the superposition of fields radiated from two elements:
$$\begin{aligned}\mathbf{E}(\mathbf{r},\theta) &= \mathbf{E}_1(\mathbf{r},\theta)+\mathbf{E}_2(\mathbf{r},\theta)\\ &= \mathbf{a}_\theta \frac{j\eta k l I_0}{4\pi r}\sin\theta e^{-j(kr_1+\frac{\beta}{2})}+\mathbf{a}_\theta \frac{j\eta k l I_1}{4\pi r}\sin\theta e^{-j(kr_2-\frac{\beta}{2})}\\ &= \underbrace{\mathbf{a}_\theta \frac{j\eta k l}{4\pi r}\sin\theta e^{-jkr}}_{\text{Element Factor}}\underbrace{\left[I_0 e^{-j(\frac{1}{2}kd\cos\theta+\frac{\beta}{2})}+I_1 e^{j(\frac{1}{2}kd\cos\theta+\frac{\beta}{2})}\right]}_{\text{Array Factor}}\end{aligned}$$

- *Element Factor* $|\mathbf{E}_0(\mathbf{r},\theta)|$ or $F(\theta)$: E field radiated from one radiator at the origin.
- *Array Factor* $AF(\theta,\phi)$: The radiation pattern of an array of elements considered to radiate isotropically. $AF(\theta)=2\cos\left(\frac{1}{2}kd\cos\theta+\frac{\beta}{2}\right)$, $0\le\theta\le\pi$ if $I_1=I_2$

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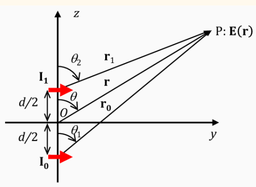
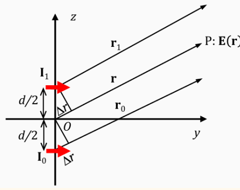


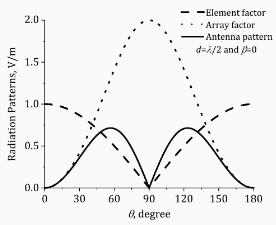


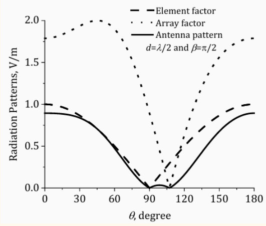
Consider an array of two identical short dipoles of length=l. For an array with an element spacing d, the excitation phase difference β between the dipoles, and the same excitation magnitude of dipoles $\mathbf{I}_0=\mathbf{a}_y I_0 e^{-j\frac{\beta}{2}}$ and $\mathbf{I}_1=\mathbf{a}_y I_1 e^{j\frac{\beta}{2}}$, find the angles of observation in y-z plane where the nulls of pattern occur. The observation point P positions in the far-field zone of the array.

Solution

1. Calculate the field $\mathbf{E}_0$ radiated from one dipole $\mathbf{I}=\mathbf{a}_y I$ located at the origin O.
   - $\mathbf{E}_0(\mathbf{r},\theta)=\mathbf{a}_\theta j\eta k I l\frac{e^{-jkr}}{4\pi r}\sin\theta$ for $\mathbf{I}=\mathbf{a}_z I$
   - $\mathbf{E}_0(\mathbf{r},\theta)=\mathbf{a}_\theta j\eta k I l\frac{e^{-jkr}}{4\pi r}|\cos\theta|$ for $\mathbf{I}=\mathbf{a}_y I$
2. Calculate total field $\mathbf{E}$ at P when $r_1\approx r_2$ for amplitude and $r-r_1=r_2-r\approx\frac{1}{2}kd\cos\theta$ for phase :
$$\begin{aligned}\mathbf{E} &= \mathbf{E}_0+\mathbf{E}_1\\ &= \mathbf{a}_\theta\frac{j\eta I_1 kl}{4\pi r_0}|\cos\theta_0|e^{-j(kr_0+\frac{\beta}{2})}+\mathbf{a}_\theta\frac{j\eta I_0 kl}{4\pi r_1}|\cos\theta_1|e^{-j(kr_1-\frac{\beta}{2})}\\ &\approx \mathbf{a}_\theta\frac{j\eta I_0 kl}{4\pi r}|\cos\theta|e^{-jkr}\,2\cos(\frac{1}{2}kd\cos\theta+\frac{\beta}{2})\\ &= \mathbf{E}_0(\mathbf{r},\theta)AF(\theta)\end{aligned}$$

array factor still applies, only the element factor is changed from sin θ to |cos θ|





Total array pattern:
$$E(\theta)=2|\cos\theta|\cos(\frac{1}{2}kd\cos\theta+\frac{\beta}{2}).\ 0\le\theta\le\pi$$
3. To find the nulls in the y-z plane,
$$E(\theta)=2|\cos\theta|\cos(\frac{1}{2}kd\cos\theta+\frac{\beta}{2})=0 \quad \text{as } \theta=\theta_{null}$$
$$\cos\theta_{null}=0 \quad \therefore\theta_{null}=\pi/2$$
and
$$\cos(\frac{1}{2}kd\cos\theta_{null}+\frac{\beta}{2})=0 \rightarrow \frac{1}{2}kd\cos\theta_{null}+\frac{\beta}{2}=\pm(\frac{2n+1}{2})\pi$$
$$\therefore\theta_{null}=\cos^{-1}[\frac{\lambda}{2\pi d}(\pm(2n+1)\pi-\beta]\quad n=0,1,2,\ldots$$

Discussion

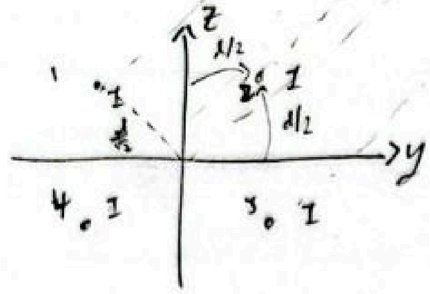
1. The null at $\theta_{null}=\pi/2$ is attributed to the radiation pattern of dipole itself.
2. Other nulls caused by AF (d and β)
3. For $\beta=0$ (in-phase excitation), the spacing d must be equal or great than half a wavelength for the existence of at least one null due to the array.

Standard Steps to Analyse Array

1. Draw the diagram: location of elements, length to fairfield (r from origin, r1, r2, …), θ is from z-axis by convention (but should always double check the definition)
2. Sum up all the far field E fields for each elements, with the corresponding element factor
3. Simplify element factor by approximating 1/r to be the same, and factorise common terms out
4. Express each of r1, r2, …, with r and θ, refer to the diagram for ± and sin(θ) or cos(θ)
5. Any phase in the current, combine it with the phase due to propagation
6. At this point, it should be of a form of element factor x array factor, where array factor is a sum of complex exponentials
7. Express sum of complex exponentials in terms of sin and cos
8. Null occurs when either element factor or array factor = 0
9. Lobe occurs when element factor x array factor is at a maxima, lobe must be between two nulls
10. **MUST REMEMBER TO GET ALL THE ANGLE FOR TRIGO SOLUTIONS**

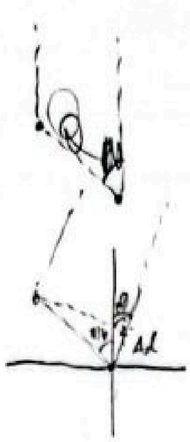
Things to Note

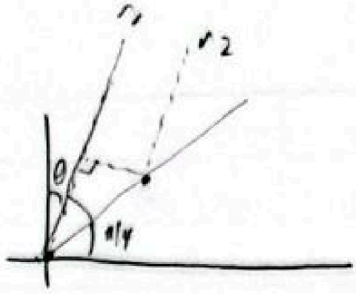
- If the current is different, if one of the element currents is very big, there could be no null, the element is dominating the field pattern. There is a certain range in current for there to be a null, solve the array factor can see.
- No matter the arrangement of elements, drawing r from origin and comparing phase of each element can always get the solution
- For a dipole along z, the EF ~ $\sin(\theta)$, $0 \le \theta \le \pi$, if its along x or y, EF ~ $|\cos(\theta)|$, $0 \le \theta \le \pi$



r_0

$r_1 =$





$|E| \sim$

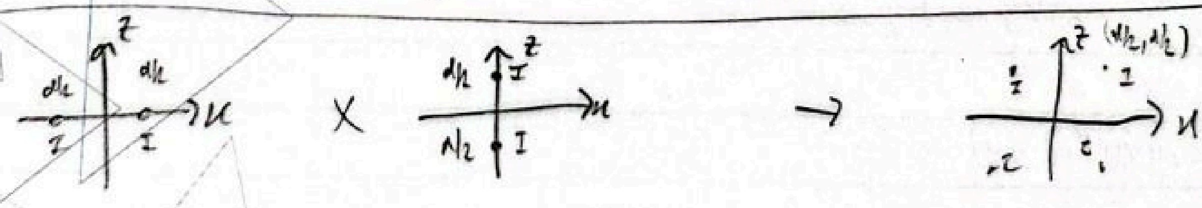
$r_1 = r_0 - \Delta l$

$\Delta l = \frac{d}{\sqrt{2}} \cos(\theta + \pi/4)$

$$r_3 = r_0 - \tfrac{d}{\sqrt{2}} \cos(\theta - \pi + \pi/4)$$
$$= r_0 + \tfrac{d}{\sqrt{2}} \cos(\theta + \pi/4)$$
$$r_4 = r_0 - \tfrac{d}{\sqrt{2}} \cos(\theta - \tfrac{3\pi}{2} + \tfrac{\pi}{4})$$
$$= r_0 + \tfrac{d}{\sqrt{2}} \cos(\theta - \pi/4)$$

$$r_1 = r_0 - \tfrac{d}{\sqrt{2}} \cos(\theta + \pi/4)$$
$$r_2 = r_0 - \tfrac{d}{\sqrt{2}} \sin(\tfrac{\pi}{2} - (\theta - \pi/4))$$
$$= r_0 - \tfrac{d}{\sqrt{2}} \sin(\tfrac{3}{4}\pi - \theta) \qquad \cos(x) = \sin(x+90)$$
$$= r_0 + \tfrac{d}{\sqrt{2}} \sin(\theta - \tfrac{3\pi}{4}) \qquad \sin x = \cos(x-90)$$
$$= r_0 + \tfrac{d}{\sqrt{2}} \cos(\theta - \tfrac{5}{4}\pi)$$
$$= r_0 + \tfrac{d}{\sqrt{2}} \cos(\theta - \pi - \tfrac{1}{4}\pi)$$
$$= r_0 - \tfrac{d}{\sqrt{2}} \cos(\theta - \tfrac{\pi}{4})$$

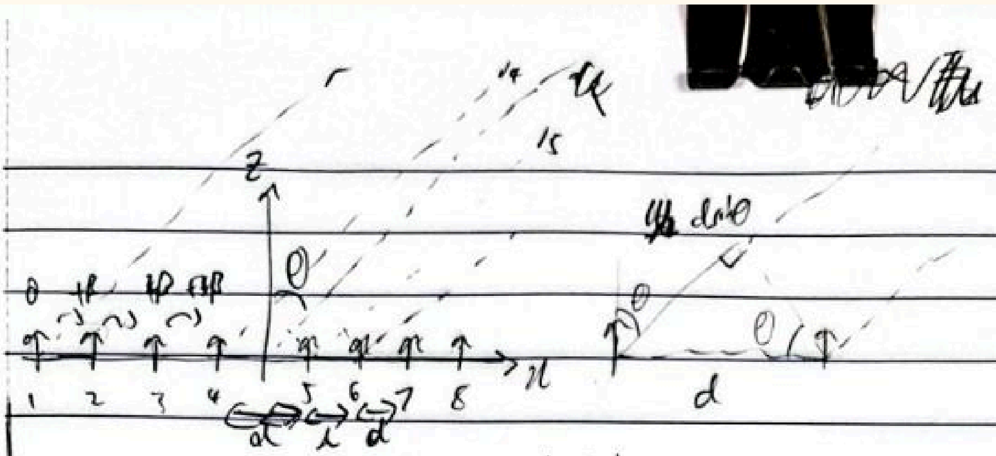
$$|E| \sim e^{-jk(r_0 - \frac{d}{\sqrt{2}}\cos(\theta+\pi/4))} + e^{-jk(r_0 - \frac{d}{\sqrt{2}}\cos(\theta-\pi/4))} + e^{-jk(r_0 + \frac{d}{\sqrt{2}}\cos(\theta+\pi/4))} + e^{-jk(r_0 + \frac{d}{\sqrt{2}}\cos(\theta-\pi/4))}$$
$$\sim e^{j(\frac{kd}{\sqrt{2}}\cos(\theta+\pi/4))} + e^{-j(\frac{kd}{\sqrt{2}}\cos(\theta+\pi/4))} + e^{j(\frac{kd}{\sqrt{2}}\cos(\theta-\pi/4))} + e^{-j(\frac{kd}{\sqrt{2}}\cos(\theta-\pi/4))}$$
$$= 2\cos(\tfrac{kd}{\sqrt{2}}\cos(\theta+\pi/4)) + 2\cos(\tfrac{kd}{\sqrt{2}}\cos(\theta-\pi/4))$$
$$= 2(\cos A + \cos B)$$
$$= 4\cos(\tfrac{A+B}{2})\cos(\tfrac{A-B}{2})$$
$$= 4\cos(\tfrac{kd}{2\sqrt{2}}[\cos(\theta+\pi/4)+\cos(\theta-\pi/4)])\cos(\tfrac{kd}{2\sqrt{2}}[\cos(\theta+\pi/4)-\cos(\theta-\pi/4)])$$
$$= 4\cos(\tfrac{kd}{\sqrt{2}}\cos(\theta)\cos(\pi/4))\cos(-\tfrac{kd}{\sqrt{2}}\sin(\theta)\sin(\pi/4))$$
$$= 4\cos(\tfrac{kd}{2}\cos\theta)\cos(\tfrac{kd}{2}\sin\theta)$$



$|E| \sim 2\cos(\frac{kd}{2}\sin\theta)$ $\qquad |E| \sim 2\cos(\frac{kd}{2}\cos\theta)$ $\qquad |E| \sim 4\cos(\frac{kd}{2}\cos\theta)\cos(\frac{kd}{2}\sin\theta)$

- array of array to skip long derivation

- phase not accounted for



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Hertzian dipole: $\bar{E} = \hat{a}_\theta j\omega\mu L I \frac{e^{-jkr}}{4\pi r}\sin\theta$

adj θ =

$\sin\theta = \frac{opp}{h}$

$opp = \sin\theta \, d$

at r: $\underset{\sim}{E} \propto I_1 \frac{e^{-j(kr_1+\beta)}}{r_1} + I_2 \frac{e^{-j(kr_2+2\beta)}}{r_2} + \dots$ $\qquad \frac{1}{r_1} \approx \frac{1}{r_2} \approx \frac{1}{r_3} \dots \approx \frac{1}{r}$

$\propto e^{-j(kr_1+\beta)} + e^{-j(kr_2+2\beta)} + \dots$ $\qquad I_1 = I_2 = \dots I_8$

$= e^{-j(kr+\beta)} + e^{-j(k(r-d\sin\theta)+2\beta)} + \dots$ $\qquad r_1 = r$

$e^{-j(k(r-2d\sin\theta)+3\beta)} + \dots + e^{-j(k(r-7d\sin\theta)+7\beta)}$ $\qquad r_2 = r - d\sin\theta$

$= e^{-jkr} \boxed{\sum_{n=0}^{7} e^{jn(kd\sin\theta+\beta)}}$ $\qquad r_3 = r - 2d\sin\theta$

$$AF = \sum_{n=0}^{7} e^{jn(kd\sin\theta+\beta)} \qquad N=8 \qquad \Sigma(a, ar, ar^2, \dots ar^n)$$

$$= \frac{1-e^{j(kd\sin\theta+\beta)\times N}}{1-e^{j(kd\sin\theta+\beta)}} \quad \text{for } e^{j\psi} \neq 1 \qquad \psi = kd\sin\theta+\beta \qquad = a\left(\frac{1-r^{n+1}}{1-r}\right)$$

$$= \frac{1-e^{j\psi N}}{1-e^{j\psi}}$$

$$= (1-e^{jN\psi})(1-e^{j\psi})^{-1} \qquad 1-e^{j\psi} \neq 0$$

$$\frac{dAF}{d\psi} = -(1-e^{jN\psi})(1-e^{j\psi})^{-2}(-je^{j\psi}) + (1-e^{j\psi})^{-1}(-jNe^{jN\psi}) = 0$$

$$(1-e^{jN\psi})(je^{j\psi}) - (1-e^{j\psi})(jNe^{jN\psi}) = 0$$

$je^{j\psi} = 0 \qquad (1-e^{jN\psi}) - (1-e^{j\psi})(Ne^{j(N-1)\psi}) = 0$

$e^{j\psi} = 0 \qquad (1-e^{jN\psi}) - N(e^{j(N-1)\psi} - e^{jN\psi}) = 0$

N/A

if $e^{j\psi} = 1$, $AF = N$

$\psi = 0$

$kd\sin\theta + \beta = 0$

$\beta = -kd\sin\theta$